

House Price Prediction Project Report

1. Introduction

Predicting house prices accurately is a fundamental real-world application in the real estate sector. This project implements an end-to-end supervised machine learning pipeline using **Linear Regression** to predict property market prices based on three core numerical features: **area** (sq. ft.), **rooms**, and **age** (years).

2. Project Architecture & Workflow

The project features a complete production-ready pipeline split into:

- **Exploratory Data Analysis (EDA):** Notebook-based exploration (`HousePricePrediction.ipynb`) analyzing data distributions, missing values, and feature correlations.
- **Model Training (`model/train_model.py`):** Standalone script to clean data, perform an 80-20 train/test split, train a Scikit-Learn Linear Regression model, and serialize it (`house_price_model.pkl`).
- **Backend REST API (`backend/app.py`):** Built with Flask and Flask-CORS to serve predictions and metadata over HTTP endpoints (`/`, `/model-info`, `/predict`).
- **Frontend Web Application (`frontend/streamlit_app.py`):** Built with Streamlit to provide an interactive user interface for instant valuations.

3. Dataset & Features

- **Dataset Source:** `data/house_price.csv` containing 300 clean records with zero missing values.
- **Features:**
 - `area`: Built-up area of the house in square feet (Range: 601 – 4,498 sq. ft.)
 - `rooms`: Total number of rooms (Range: 1 – 6 rooms)
 - `age`: Age of the property in years (Range: 0 – 29 years)
 - `price`: Market price of the house (Target variable; Mean: ₹6,42,081)

4. Model Training & Evaluation

The dataset was split into an 80% training set and a 20% testing set. A standard Linear Regression model was fitted using Scikit-Learn.

Evaluation Metrics:

- **Mean Absolute Error (MAE):** $\approx 116,635$
- **Root Mean Squared Error (RMSE):** $\approx 141,389$
- **R^2 Score:** ≈ 0.79

An R^2 score of **0.79** indicates that approximately **79% of the variance** in house prices is successfully explained by the three given features, representing a strong predictive baseline.

5. Conclusion & Future Scope

- **Conclusion:** The project successfully demonstrates a complete machine learning workflow ranging from exploratory data analysis and model training to robust full-stack deployment via Flask and Streamlit.
- **Future Improvements:**
 - Incorporate categorical and advanced spatial features (e.g., location, neighborhood safety index, proximity to city centers).
 - Experiment with advanced non-linear regression models such as Random Forest Regressor or XGBoost to lower prediction error.